Framework of the agent-based model for relocation scenarios

START

Agents are generated in the flooded area per group

Are **pixels available** in the same group as the agent?

Yes

Are there **transport** and **infrastructure** variables **enabled**

Yes

Are the **expansion settings** enabled?

Yes

Is the **aggressive urban expansion** scenario enabled?

Yes

Agents move to areas with similar groups and the highest infrastructure and transport value under a **aggressive urban expansion scenario.**

No

Agents **don't move** the original point.

Scenario 1

No

Agents **move** to areas with **similar group** properties randomly.

Scenario 2

No

Agents move to areas with similar groups and the **highest infrastructure and transport** value only at the urban perimeter.

Scenario 3

No

Agents move to areas with similar groups and the highest infrastructure and transport value under a **strict urban expansion scenario.**

Scenario 4

Scenario 5

Begin and end points

Parameter

Initial / final outcomes